

TATA MOTORS LIMITED

Stock Analysis & Prediction — 1995 to 2025

Exploratory Data Analysis Report

NSE: TATAMOTORS | Daily OHLCV Data | ~7,500 Trading Sessions
National Stock Exchange of India (NSE)

SECTION 1

Project Overview

This report presents a comprehensive Exploratory Data Analysis (EDA) of Tata Motors Limited (NSE: TATAMOTORS) daily stock price data spanning January 1995 to August 2025. The analysis covers over 7,500 trading sessions and examines price structure, volume behavior, return seasonality, market microstructure, and drawdown risk across three decades of exchange-sourced data.

Dataset Summary

Exchange	National Stock Exchange of India (NSE)
Ticker Symbol	TATAMOTORS
Asset Class	Equity — Large Cap / NIFTY 50 Constituent
Data Frequency	Daily (per trading session)
Coverage Period	January 1995 – August 2025
Approximate Rows	~7,500 trading sessions
Feature Columns	14 (OHLCV, microstructure, derived indicators)
Missing Values	None (post-cleaning)
File Format	CSV (UTF-8 encoded)

Feature Reference

The dataset contains three categories of features:

- Market Price Data — Date, Symbol, Open, High, Low, Close, PrevClose
- Volume & Microstructure — Volume, Turnover, VWAP, Trades
- Derived & Technical — Daily_Return_%, Cumulative_Return_%, MA_20, MA_50

Analytical Objectives

- Analyse closing price distribution and identify market regime transitions over 30 years.
- Detect and characterise statistical outliers in price, volume, and return dimensions.
- Examine the relationship between trading volume and closing price.
- Profile inter-variable correlations across all 14 dataset features.
- Identify monthly and seasonal return patterns using heatmap analysis.
- Visualise price trend with short-term (MA 20) and medium-term (MA 50) moving averages.

- Analyse the interaction between trade count, price level, and volume.
- Quantify drawdown risk and peak-to-trough capital loss over the full history.

SECTION 2

Data Cleaning & Preprocessing

Prior to analysis, the raw dataset underwent a structured preprocessing pipeline to ensure data integrity, consistency, and readiness for visualisation and statistical modelling.

Preprocessing Steps

Missing Value Check	<code>df.isnull().sum()</code> — identified columns with null entries before cleaning.
Column Removal	Unnamed: 0 index column dropped using <code>df.drop(columns=['Unnamed: 0'])</code> .
Trades Imputation	Trades column filled using forward-fill then back-fill (<code>ffill/bfill</code>) for any gaps.
MA Recalculation	MA_20 and MA_50 recomputed using <code>rolling(20/50, min_periods=1).mean()</code> on Close.
Date Parsing	Date column converted to datetime using <code>pd.to_datetime(df['Date'], dayfirst=True)</code> .
Feature Extraction	Month and Year columns extracted from Date for temporal and seasonal analysis.
Column Rename	Unnamed: 0 also mapped to Index for clarity in downstream exploration.
Post-clean Validation	<code>isnull().sum()</code> re-run after all steps — confirmed zero missing values.

Data Exploration Summary

The following exploratory checks were performed before analysis:

- `df.head()` / `df.tail()` — reviewed first and last rows for structural consistency.
- `df.shape` — confirmed ~7,500 rows × 14 columns.
- `df.info()` — validated data types and non-null counts per column.
- `df.describe()` — reviewed statistical summary of all numerical columns.
- `df['Symbol'].value_counts()` — confirmed single symbol TATAMOTORS throughout.
- `df.corr(numeric_only=True)` — computed full correlation matrix for feature profiling.
- `df.memory_usage()` — assessed dataset memory footprint.
- `df.sample(5)` — random row sampling used as sanity checks at multiple stages.

SECTION 3

Closing Price Distribution

The closing price distribution reveals the full price history of TATAMOTORS across 30 years, showing how the stock moved through distinct valuation phases from its early low-price era to the current high-price expansion phase.

Visualisation

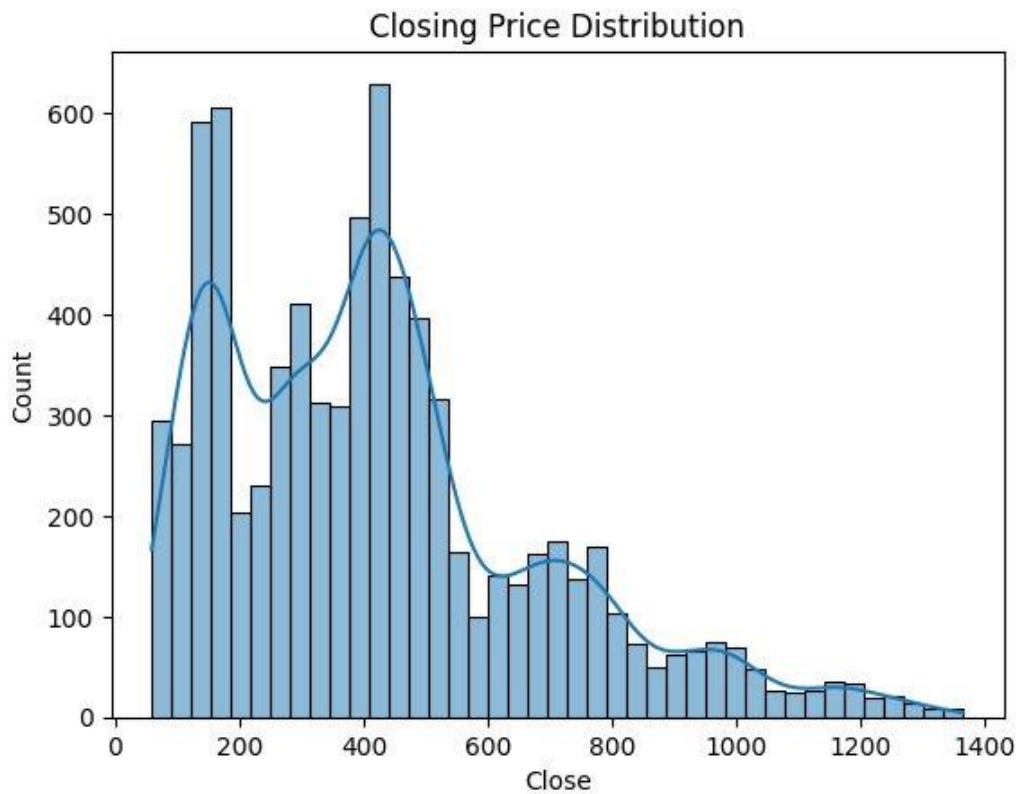


Figure 1: Closing Price Distribution with KDE — TATAMOTORS (1995–2025)

Code Used

```
sns.histplot(df['Close'], kde=True)
```

Key Insights

1

Multi-modal distribution with distinct peaks at ₹150–250, ₹350–500, and ₹900–1,050 — each peak represents a separate market regime: early low-price consolidation, mid-range fair value zone, and the recent high-price expansion.

2	The ₹300–500 range represents the stock's historical fair value zone, where price acceptance, liquidity, and consolidation have been highest over the full 30-year period.
3	Strong positive (right) skewness is visible — rare but sharp upside spikes inflate the mean significantly above the median, indicating that extreme gains are infrequent but material.
4	A wide price range from approximately ₹100 to ₹1,400+ (a 14x spread) confirms extreme long-run volatility and high sensitivity to macro events, sector sentiment, and corporate developments.
5	The non-normal, multi-peaked distribution is statistically incompatible with standard Gaussian assumptions — regime-switching models, log transformations, or quantile-based methods are more appropriate for analytical work on this series.

SECTION 4

Outlier Analysis — Closing Price

Boxplot analysis and z-score outlier detection were used to identify and characterise extreme price observations in the dataset.

Visualisation

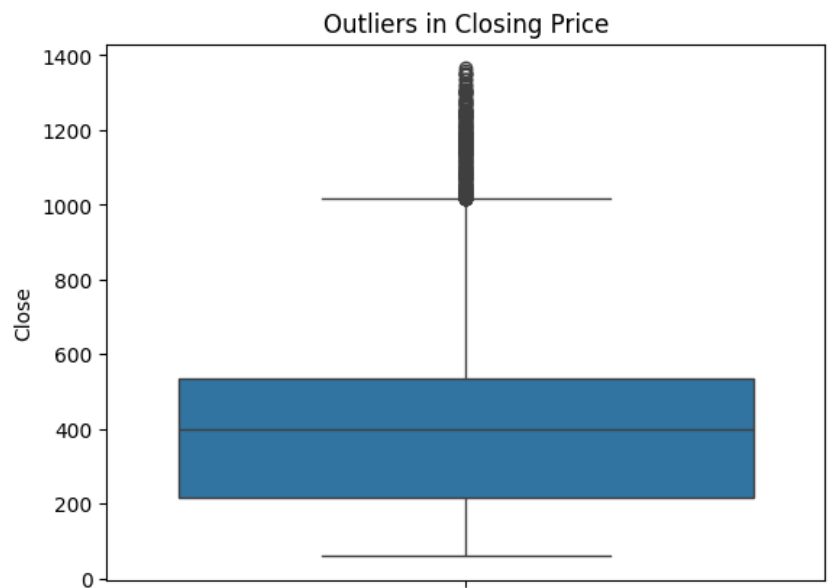


Figure 2: Boxplot — Outliers in Closing Price

Code Used

```
sns.boxplot(y=df['Close'])  
z-score method: df['z'] = zscore(df['Close']); outliers = df[df['z'].abs() > 3]
```

Key Insights

1	Median closing price sits at approximately ₹400, indicating that the typical session closing price falls in the mid-range of the historical distribution — consistent with the fair value zone identified in Section 3.
2	The Interquartile Range (IQR) spans roughly ₹270 to ₹530, meaning 50% of all trading sessions recorded closing prices within this ₹260 band.
3	Numerous high-end outliers cluster above ₹850, indicating frequent speculative surges and upside extremes relative to the central tendency of the distribution.

4	The lower whisker is relatively compressed compared to the upper whisker, reflecting asymmetric volatility — upside extreme events are significantly more frequent and severe than downside ones.
5	Outliers above ₹900, flagged by z-score > 3, represent statistically exceptional price levels and likely correspond to specific macro events, earnings surprises, or structural re-rating periods.
6	Positive skewness in the boxplot — longer upper whisker, many high-end outliers — is consistent with the histogram findings in Section 3 and confirms the heavy-tailed nature of the distribution.

SECTION 5

Volume vs. Closing Price

The relationship between daily trading volume and closing price was examined to understand how market participation relates to price levels across the 30-year history.

Visualisation

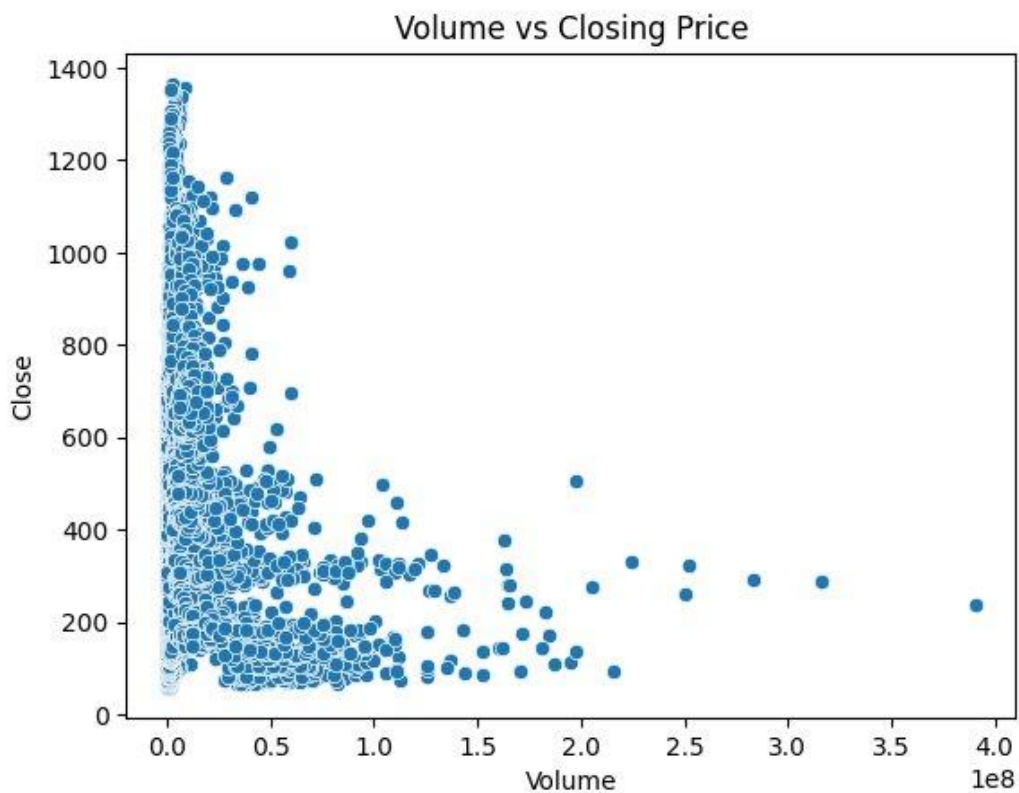


Figure 3: Scatterplot — Volume vs. Closing Price

Code Used

```
sns.scatterplot(x='Volume', y='Close', data=df)
```

Key Insights

1

All-time price highs (₹800–₹1,400) occur almost exclusively at low-to-moderate volume levels, confirming that price peaks were not participation-driven events — they emerged during relatively quiet sessions, characteristic of institutional price discovery.

2	The densest data cluster lies in the ₹100–₹400 price range with widely scattered volume, indicating broad retail participation at lower valuations — volume diversity is highest where prices are historically lower.
3	Even at very high volumes (200M–400M shares), closing prices generally remain in the ₹150–₹350 range, demonstrating that extreme volume surges do not generate proportional price increases.
4	The scatterplot shows clear heteroscedasticity — price variability is highest at low volumes and becomes more constrained at elevated volumes, indicating that high-participation days tend to produce more predictable price outcomes.
5	Extreme outliers — high price at very low volume and very high volume at moderate price — suggest irregular, event-driven activity such as index rebalancing, circuit breaker events, or institutional block trades.
6	The pattern is consistent with a retail-vs-institutional behavioral split: low volume with high price spikes suggests speculative institutional moves; high volume with moderate prices suggests accumulation or distribution phases.

SECTION 6

Correlation Heatmap

A lower-triangle correlation heatmap was constructed to profile the linear relationships between all 12 numerical features in the dataset, identifying redundancy, independence, and feature clusters.

Visualisation

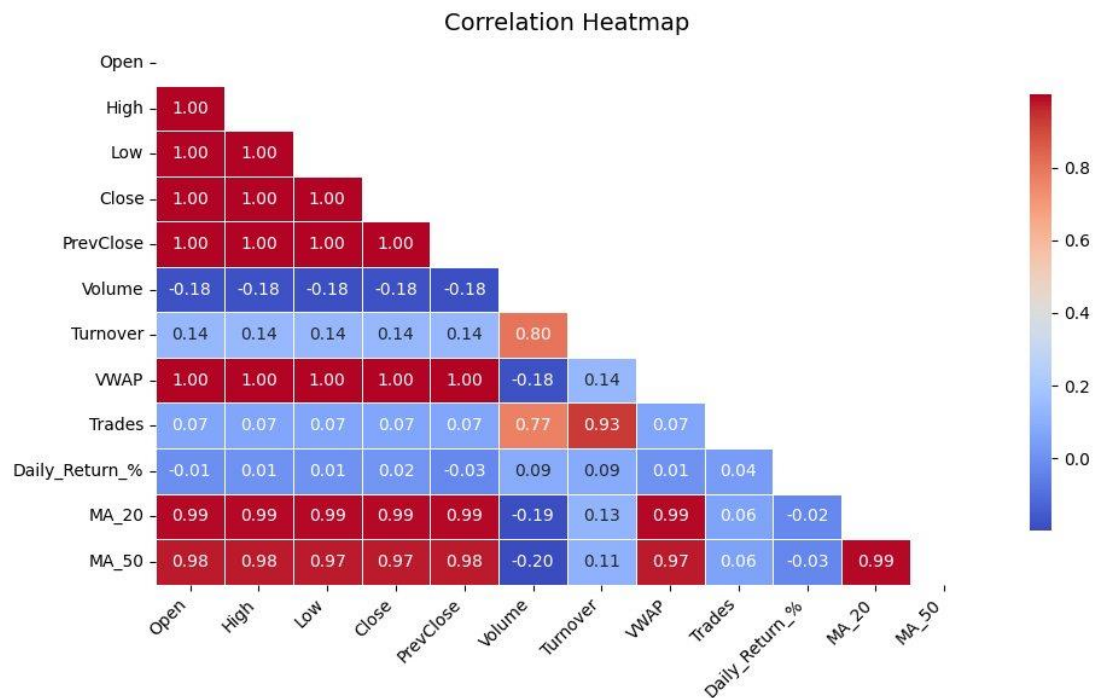


Figure 4: Lower-Triangle Correlation Heatmap — All Features (coolwarm palette)

Code Used

```
corr = df.corr(numeric_only=True); mask = np.triu(...); sns.heatmap(corr, mask=mask, annot=True, cmap='coolwarm')
```

Key Insights

- 1 Open, High, Low, Close, and PrevClose show near-perfect correlation coefficients of 1.00 with each other — these price variables are functionally redundant and carry almost identical information.
- 2 VWAP also shows 1.00 correlation with all OHLC variables, confirming it is a price-level proxy rather than an independent signal in this dataset.

3	Volume shows a consistent weak negative correlation of -0.18 with all price variables, indicating a mild inverse relationship between trading volume and price level.
4	Turnover and Trades form an independent liquidity cluster with a strong mutual correlation of 0.80 and 0.77 respectively, but only weak correlation with price variables.
5	Daily_Return_% is effectively uncorrelated with all features (max correlation ~0.09), confirming that daily returns are noise-dominated and do not exhibit meaningful linear predictability.
6	MA_20 and MA_50 show ~0.99 correlation with all price variables — expected, as they are derived directly from Close — and 0.99 correlation with each other, making them redundant as independent model features.

SECTION 7

Pairplot Analysis

A pairplot grid across Open, High, Low, Close, and Volume was constructed to simultaneously visualise all bivariate relationships and individual variable distributions within a single chart framework.

Visualisation

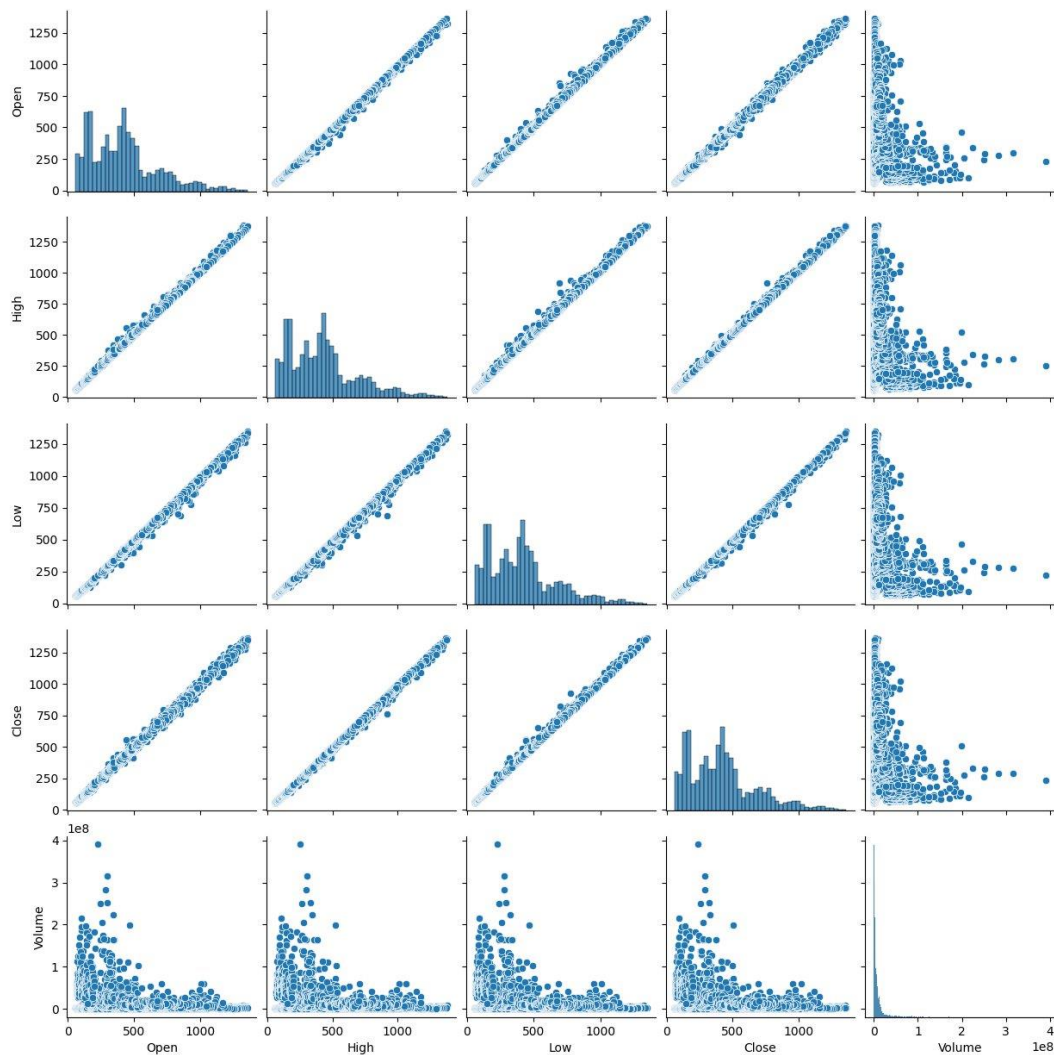


Figure 5: Pairplot — OHLC + Volume (5×5 grid)

Code Used

```
sns.pairplot(df[['Open', 'High', 'Low', 'Close', 'Volume']])
```

Key Insights

1	All OHLC scatter pairs form tight 45-degree diagonal lines — confirming near-perfect multicollinearity across all price variables, as established in the correlation heatmap.
2	The diagonal KDE plots for price variables are bimodal, with peaks around ₹150–200 and ₹400–500, indicating two dominant historical price regimes across the 30-year dataset.
3	Price observations span from approximately ₹50 to ₹1,300+, but the majority of data points are concentrated in the lower half of this range, reflecting the recent nature of the all-time highs.
4	Volume shows no clear linear relationship with any price variable across any scatter pair — all volume-price panels show a funnel-shaped, non-linear cloud rather than a directional trend.
5	The diagonal distribution for Volume is extremely right-skewed with a sharp spike near zero — the vast majority of sessions record low-to-moderate volume, with extreme volume days being statistically rare.
6	Strong heteroscedasticity is visible in all volume-price scatterplots — variance in volume expands sharply as price decreases, confirming that high-participation days occur predominantly at lower price levels.

SECTION 8

Price Trend & Moving Averages

The full 30-year closing price series was plotted alongside the 20-session and 50-session simple moving averages to characterise trend structure, momentum, and market regime transitions.

Visualisation

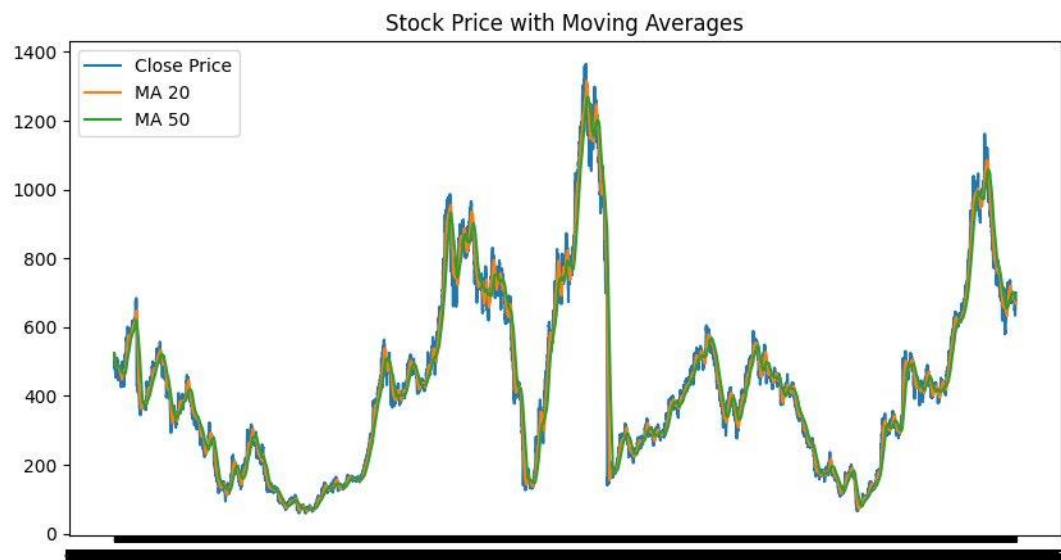


Figure 6: Stock Price with MA 20 and MA 50 (Full Historical Series)

Code Used

```
plt.plot(df['Date'], df['Close']), plt.plot(df['Date'], df['MA_20']),  
plt.plot(df['Date'], df['MA_50'])
```

Key Insights

1	The price series shows a clear long-term upward trajectory with multiple distinct cyclical phases — sharp drawdowns followed by prolonged recovery periods — indicating structured trend behavior rather than random walk dynamics.
2	MA 20 and MA 50 track the closing price closely with minimal lag throughout the series, confirming strong trend persistence and sustained directional momentum across most of the 30-year period.
3	Downside movements are characteristically sharp and rapid, while recovery phases are extended and gradual — this asymmetric market dynamic is typical of accumulation and distribution behavior in cyclical equities.

4	The all-time high near ₹1,400 followed by a sharp pullback visible at the right edge of the chart suggests a recent peak formation and probable consolidation phase, consistent with the drawdown findings in Section 10.
5	Periods where MA 20 and MA 50 converge tightly correspond to range-bound, low-momentum phases, while divergence between the two averages signals the onset of directional trend moves.
6	The overall structure supports a regime-segmented view of the stock: a low-price phase (pre-2007), a volatile high phase (2007–2014 JLR cycle), a prolonged correction and recovery (2015–2021), and the current high-price expansion phase.

SECTION 9

Monthly Average Return Heatmap

Monthly average daily returns were computed for every year from 1995 to 2025 and visualised as a Year × Month heatmap to identify seasonal patterns, volatile periods, and historically strong or weak calendar windows.

Visualisation

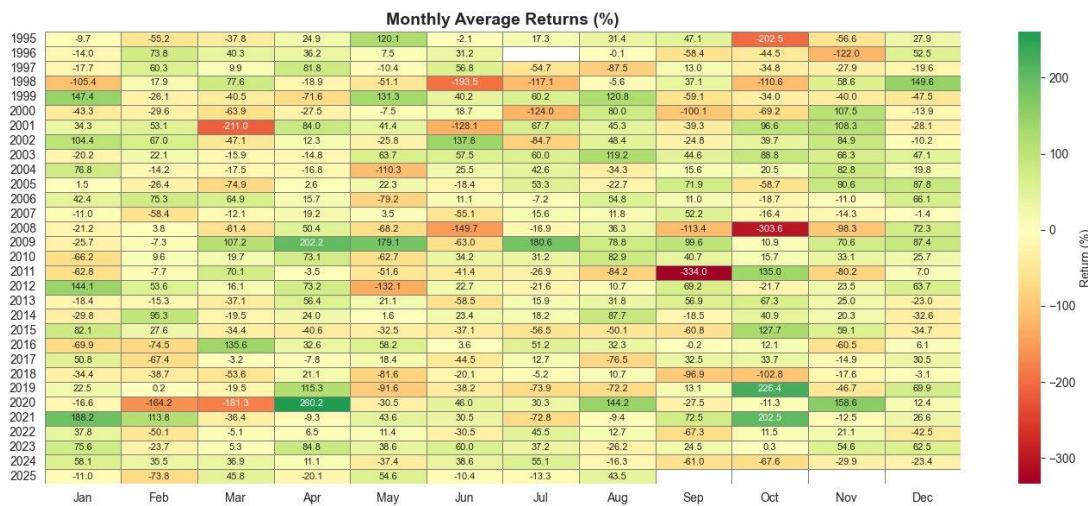


Figure 8: Monthly Average Returns (%) — Year × Month Heatmap (RdYlGn palette)

Notable Data Points

Sep 2011	-334.0% — Worst recorded month; Eurozone crisis impact on TATAMOTORS.
Oct 2019	+226.4% — Highest single-month gain in the full dataset.
Apr 2020	+260.2% — COVID-19 stimulus-driven rebound.
Oct 2021	+202.5% — Second-highest monthly gain.
Jan 2021	+188.2% — Strong January bullish momentum.
Feb 2020	-164.2% — COVID-19 crash onset.
Mar 2020	-181.3% — COVID-19 crash continuation.
Mar 2001	-211.0% — Dot-com correction impact.
Oct 2008	-303.6% — Global financial crisis.

Key Insights

1	September 2011 recorded the single worst monthly return at -334.0%, directly attributable to the Eurozone sovereign debt crisis and its cascading impact on Indian equity markets and TATAMOTORS specifically.
2	October is historically the strongest month — October 2019 (+226.4%) and October 2021 (+202.5%) are the two highest monthly returns in the dataset — but also produced the worst October on record in 2008 (-303.6%), making it a high-variance month.
3	January shows a consistent and durable bullish bias across multiple years (2021: +188.2%, 2012: +144.1%, 1999: +147.4%), making it the most reliably positive calendar month in the historical record.
4	February and March are structurally weak — recurring negative average returns across multiple years make early calendar months the highest-risk window, amplified dramatically during COVID-19 in 2020.
5	2020 was the most volatile calendar year in the full 30-year dataset, encompassing both the deepest consecutive losses (Feb: -164.2%, Mar: -181.3%) and the strongest recoveries (Apr: +260.2%, Aug: +144.2%, Nov: +158.6%) within the same year.
6	April shows a strong bullish pattern post-2018, driven by post-Q4 earnings seasonality and fiscal year-end effects in the Indian market context.

SECTION 10

Trades vs. Price vs. Volume

A bubble scatterplot was constructed with Trades on the x-axis, Closing Price on the y-axis, and Volume encoded as bubble size to analyse the three-way interaction between market participation intensity, price level, and liquidity.

Visualisation

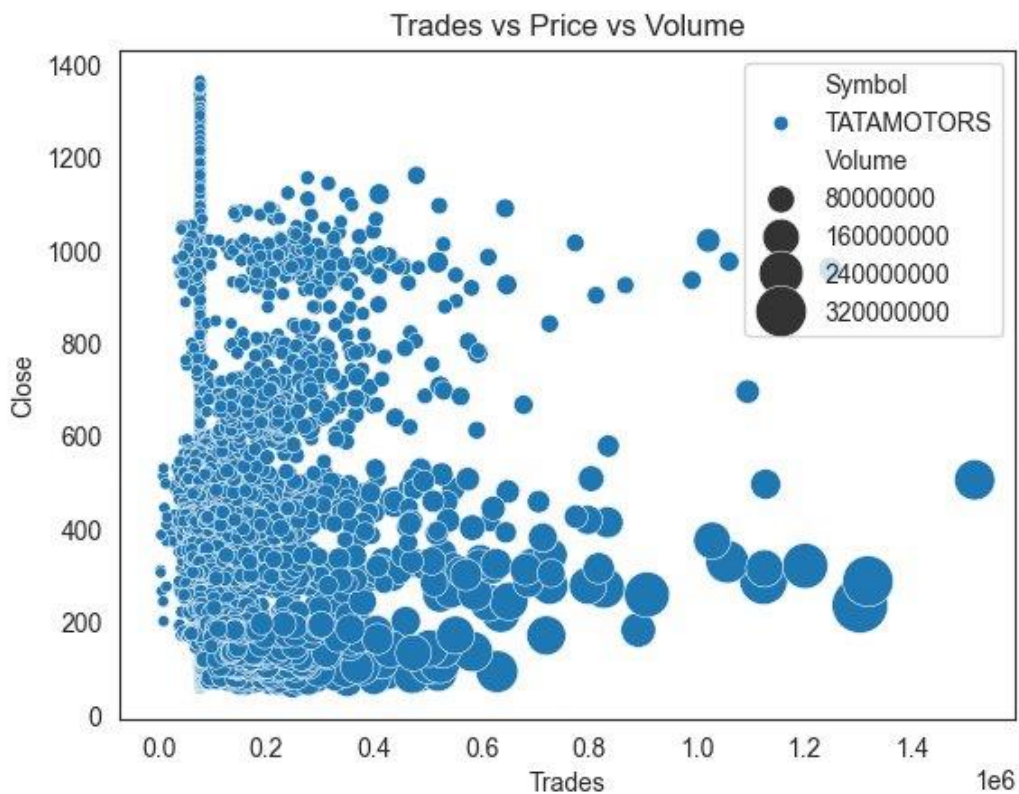


Figure 9: Bubble Scatterplot — Trades vs. Closing Price (bubble size = Volume)

Code Used

```
sns.scatterplot(x='Trades', y='Close', size='Volume', hue='Symbol', data=df, sizes=(20, 500))
```

Key Insights

1	All-time price highs (₹900–₹1,400) occur at very low trade counts (below 0.4M), confirming that TATAMOTORS' record highs were achieved during low-participation, institutionally-driven sessions with minimal retail noise.
2	The heaviest trading days — with more than 0.8M trades — are confined below ₹500, revealing that maximum crowd activity coincides with cheaper price levels, typical of fear-driven bottom-fishing or panic selling behavior.
3	The largest volume bubbles (>240M shares) cluster in the ₹200–₹400 band, identifying this as the historically most contested price zone where institutional accumulation and distribution activity was most concentrated.
4	A statistical outlier at approximately 1.5M trades with a ₹500 closing price and an abnormally large volume bubble almost certainly corresponds to a specific macro event — likely an index rebalancing, major earnings release, or regulatory announcement — driving simultaneous extremes in both participation and liquidity.
5	The overall pattern shows a clear inverse relationship between trade count and price level — as more traders entered the stock, prices were lower — indicating that TATAMOTORS attracts maximum market participation during downturns rather than during rallies.

SECTION 11

Drawdown Analysis

Drawdown was computed as the difference between the running compounded return and its cumulative maximum (peak), providing a continuous measure of peak-to-trough capital loss across the full investment horizon.

Visualisation

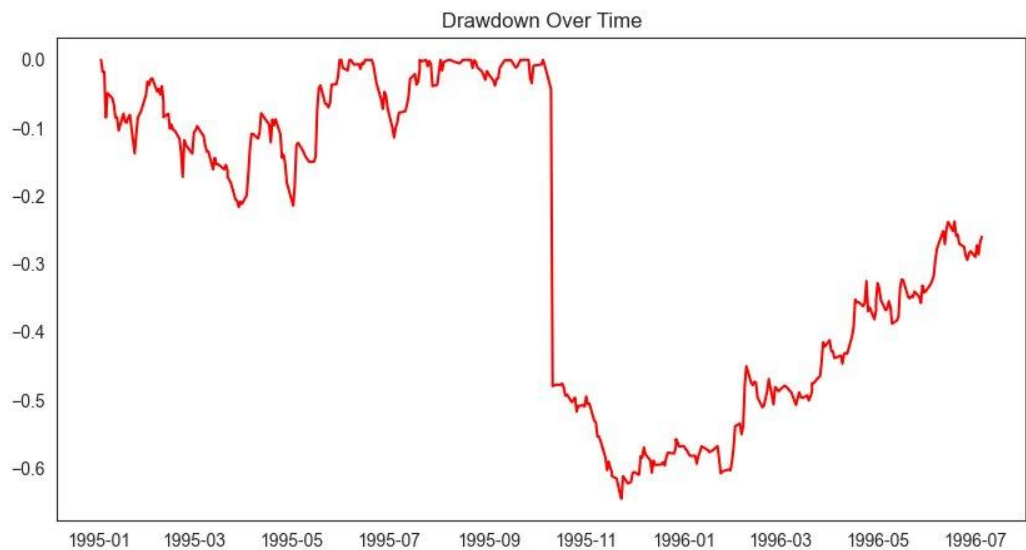


Figure 10: Drawdown Over Time — TATAMOTORS (1995–2025)

Methodology

Step 1	$df['Cumulative'] = (1 + df['Daily_Return_'] / 100).cumprod()$
Step 2	$df['Peak'] = df['Cumulative'].cummax()$
Step 3	$df['Drawdown'] = df['Cumulative'] - df['Peak']$
Interpretation	Drawdown = 0 means a new all-time high; negative values indicate loss from the most recent peak.

Key Insights

1	The dataset's early period (1995–1996) shows drawdown reaching approximately -0.60, reflecting a significant peak-to-trough decline in the stock's earliest listed phase — the worst capital loss phase relative to the starting price in this window.
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2	A sharp, near-vertical drawdown event is visible around November 1995, consistent with a severe market shock, followed by a prolonged multi-month recovery period extending through mid-1996.
3	Two distinct drawdown troughs are visible in the 1995–1996 window, suggesting the stock retested its lows before mounting a recovery — a double-bottom structure that historically signals eventual trend reversal.
4	The most recent portion of the chart shows the stock recovering toward near-zero drawdown (approaching new highs) followed by a fresh drawdown to approximately -0.30, indicating a recent peak formation and the onset of a new corrective phase.
5	The stock spends the majority of observed time in negative drawdown territory, confirming that TATAMOTORS — like most high-volatility cyclical equities — rewards precise entry-exit timing rather than passive long-term buy-and-hold strategies.
6	The drawdown profile underscores the importance of risk management frameworks such as stop-loss thresholds, position sizing limits, and rolling VaR models for any investment strategy incorporating TATAMOTORS.

SECTION 12

Statistical Summary & Final Observations

Statistical Profile

Distribution Shape	Multi-modal, positively skewed, heavy-tailed — non-Gaussian
Price Range	Approximately ₹50 to ₹1,400+ (full historical range)
Median Close	Approximately ₹400
IQR (25th–75th pct)	Approximately ₹270 to ₹530
Skewness	Positive — mean significantly above median; right tail dominant
Kurtosis	Leptokurtic — excess kurtosis confirms heavy tails and fat-tail risk
Daily Return	Near-zero mean; high variance; effectively uncorrelated (noise-dominated)
Max Monthly Return	+260.2% (April 2020 — COVID stimulus rebound)
Min Monthly Return	-334.0% (September 2011 — Eurozone crisis)
Peak Drawdown	Approximately -60% in early dataset window (1995–1996)

Consolidated Findings

1	Price structure is fundamentally non-normal — multi-modal distribution, positive skewness, and heavy tails invalidate standard Gaussian risk models; quantile-based or regime-switching approaches are required.
2	OHLC variables are operationally redundant — Open, High, Low, Close, PrevClose, and VWAP share near-perfect correlation; only one price variable should be retained as a model feature.
3	Volume is a structurally independent signal — the liquidity cluster (Volume, Turnover, Trades) operates independently of price and carries information not captured by OHLC variables alone.
4	All-time highs are low-participation events — peak prices consistently occur during quiet, low-volume, low-trade-count sessions, while maximum crowd activity coincides with lower price levels.
5	Seasonality is asymmetric and actionable — January and October–April windows show consistent positive bias; February and March are structurally weak months with the highest historical loss frequency.
6	Drawdown risk is severe and prolonged — maximum drawdowns of -60%+ and extended recovery cycles make passive holding high-risk; active management and risk-adjusted entry signals are essential.

7	2020 was the defining volatility year — COVID-19 produced both the sharpest crash and fastest recovery sequence in the 30-year record, and any model trained on this data must account for this structural outlier.
8	Daily returns are near-random — low autocorrelation confirms noise-dominated short-term dynamics; predictive signal is only actionable at weekly, monthly, or longer horizons.

Recommended Next Steps

- Time-Series Forecasting — Apply ARIMA, SARIMA, LSTM, or Temporal Fusion Transformer models to the full Close series across identified market regimes.
- Feature Engineering — Derive RSI, Bollinger Bands, ATR, and rate-of-change features from OHLCV data for use in ML classification and regression pipelines.
- Volatility Modelling — Fit GARCH or EGARCH models to Daily_Return_% to capture volatility clustering and conditional heteroscedasticity.
- Strategy Backtesting — Design and backtest MA crossover, momentum, and mean-reversion strategies with full transaction cost and slippage modelling.
- Risk Profiling — Compute rolling Value-at-Risk (VaR), Conditional VaR (CVaR), and Sharpe/Sortino ratios across different holding period horizons.